

Name: \_\_\_\_\_

# TRANSFORMATIONS, (ENLARGEMENT)

Date: \_\_\_\_\_

**LO: I can enlarge a shape by a given scale factor from a centre of enlargement.**

## Enlargement

An **enlargement** changes the size of a shape using a **scale factor** and a **centre of enlargement**. Multiply each distance from the centre by the scale factor.

**Note:** SF > 1 makes bigger.  $0 < \text{SF} < 1$  makes smaller. Negative SF gives enlargement on the other side.

## Quick examples

|                                                                                   |                                                |                             |
|-----------------------------------------------------------------------------------|------------------------------------------------|-----------------------------|
|  | SF 2 from centre: all distances doubled        | basic enlargement           |
|  | SF $\frac{1}{2}$ : shape becomes half the size | fractional SF               |
|  | SF -1: same size, rotated 180°                 | negative SF                 |
|  | Enlargement only makes shapes bigger           | fractional SF makes smaller |

## Key idea

New distance from centre = original distance  $\times$  scale factor.  
Count from centre to each vertex.

$$\text{New position} = \text{Centre} + \text{SF} \times (\text{Point} - \text{Centre})$$

Multiply the vector from centre to point by SF

## MODELLED EXAMPLES

Study each example carefully before attempting the practice questions.

### Example 1 – SF 2 from origin

Enlarge triangle with vertices (1,1), (3,1), (1,3) by SF 2, centre (0,0).

Multiply each coordinate by 2  
(1,1) (2,2), (3,1) (6,2), (1,3) (2,6)  
New triangle: (2,2), (6,2), (2,6)

**(2,2), (6,2), (2,6)**

When centre is origin, just multiply all coordinates by SF.

### Example 2 – SF 3 from a point

Enlarge point (2,1) by SF 3, centre (1,0).

Vector from centre to point: (2-1, 1-0) = (1, 1)  
Multiply by 3: (3, 3)  
New position: (1+3, 0+3) = (4, 3)

**(4, 3)**

Find vector from centre, multiply by SF, add back to centre.

### Example 3 – fractional SF

Enlarge point (6,4) by SF  $\frac{1}{2}$ , centre (0,0).

Multiply coordinates by  $\frac{1}{2}$   
( $6 \times \frac{1}{2}$ ,  $4 \times \frac{1}{2}$ )  
= (3, 2)

**(3, 2)**

A scale factor less than 1 reduces the shape.

### Example 4 – finding the SF

A shape has vertex at (2,3). After enlargement from (0,0) it moves to (6,9). Find SF.

SF = new distance / original distance  
 $\frac{6}{2} = 3$  or  $\frac{9}{3} = 3$   
Scale factor = 3

**SF = 3**

Divide the new coordinate by the original.

## YOUR TURN – PRACTICE (deliberate practice)

### 1. SF from origin

Enlarge each point by the given SF from (0,0).

- a) (2, 3) SF 2
- b) (1, 4) SF 3
- c) (4, 6) SF  $\frac{1}{2}$
- d) (5, 2) SF 4

### 2. SF from a centre

Enlarge each point by SF 2 from the given centre.

- a) Point (3,2), centre (1,1)
- b) Point (4,3), centre (2,1)
- c) Point (5,4), centre (1,2)
- d) Point (2,5), centre (0,1)

### 3. Fractional scale factors

Enlarge each point from (0,0).

- a) (8, 6) SF  $\frac{1}{2}$
- b) (9, 12) SF  $\frac{1}{3}$
- c) (10, 4) SF  $\frac{1}{2}$
- d) (6, 15) SF  $\frac{2}{3}$

### 4. Finding scale factors

Find the scale factor of each enlargement from (0,0).

- a) (2,1) (6,3)
- b) (4,3) (8,6)
- c) (6,9) (2,3)
- d) (5,10) (10,20)

### 5. Mixed enlargement

Solve each problem.

- a) Enlarge (1,2) SF 3, centre (0,0)
- b) Enlarge (4,1) SF 2, centre (2,1)
- c) Shape area = 5 cm<sup>2</sup>, SF = 3. New area?
- d) Side 4 cm becomes 12 cm. Scale factor?
- e) Enlarge (3,5) SF -1, centre (1,2)

## COMMON MISCONCEPTIONS – SPOT THE MISTAKE

Each statement shows a student's answer. Tick ( ) the ones that are correct. If it is wrong, write the correct answer.

a) In an enlargement, multiply distances from the centre by the SF ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

b) A scale factor of  $\frac{1}{2}$  makes the shape bigger ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

c) Area scale factor = (length SF)<sup>2</sup> ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

d) Enlargement always increases the size of a shape ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

e) A negative scale factor places the image on the other side of the centre ☐

Correct answer: \_\_\_\_\_

Reason: \_\_\_\_\_

## 6. CHALLENGE – WORD PROBLEMS (apply your skills)

a) A triangle has vertices at (1,1), (4,1), (1,3). It is enlarged by SF 2 from centre (1,1). Find the vertices of the enlarged triangle.

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

b) A shape is enlarged by SF -2 from centre (3,2). The image of point P is at (-1, 8). Find the original position of P.

Expression: \_\_\_\_\_

Simplified: \_\_\_\_\_

### TOP TIPS

Read the question carefully. Show all your working step by step.  
Check your answer makes sense.

### CHECK YOUR WORK

Have I shown all my working clearly?

Did I check my answer using a different method?

Does my answer look reasonable?

